Project Report
On
# Crime location analysis

UNITED UNIVERSITY
PRAYAGRAJ

FACULTY OF COMPUTER APPLICATIONS

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# Introduction

## In this project

Crime analysis is the process of studying crime data to identify patterns, trends, and unsafe areas. In recent years, crime rates have increased in many locations, making public safety an important concern.

"Crime Hotspot Detection Using Data Visualization," focuses on analyzing crime data based on location, crime type, and time. The main aim of the project is to identify crime-prone areas.

# 1. Problem Statement

Crime incidents are increasing in many urban and semi-urban areas, making public safety a major concern. Traditional crime records are difficult to interpret because large datasets cannot easily reveal patterns, hotspots, or trends.

This project aims to analyze crime data using data visualization techniques to identify:

crime-prone areas,

common crime categories,

peak crime timings,

and geographic crime patterns.

# 2. Objectives of the Project

The main objectives of this project are:

To analyze crime records from different locations

To identify crime hotspot areas

To visualize crime trends over time

To compare different crime categories

To study the relationship between location, time, and crime frequency

To create interactive visualizations for better understanding

To generate insights that may help improve safety measures

# 3. Step 1 – Load Crime Dataset (CSV Database)

```
pip install notebook
```

```
pip install pandas
pip install matplotlib
pip install seaborn
pip install plotly
pip install folium
```

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
df = pd.read_csv("crime_data.csv")
```

# Load Dataset

```
df = pd.read_csv("crime_data.csv")
df.head()
df.info()
df.describe()
df.isnull().sum()
df.dropna(inplace=True)
df.drop_duplicates(inplace=True)
df['Date'] = pd.to_datetime(df['Date'])
```

# Datasheet of crime

| | |
|---|---|
| 1. Column Name. | Description |
| 2. Crime_ID- | Unique crime identifier |
| 3. Crime_Type- | Theft, Assault, obbery, etc. |
| 4. Location- | Area where crime occurred |
| 5. Date- | Crime occurrence date |
| 6. Time- | Crime occurrence time |
| 7. Latitude- | Geographic coordinate |
| 8. Longitude- . | Geographic coordinate |
| 9. STATUS- | SOLVED/UNSOLVED |

# 4. Data Cleaning & Preprocessing

A. Removed missing values

B. Eliminated duplicate records

C. Converted date and time formats

D. Standardized location names

E. Categorized crime types properly

# 5. Step 2 – Data Exploration

**In this step, the dataset is explored to understand patterns and distributions.**

**Analysis Performed:**

a.) Total number of crimes

b.) Most common crime types

c.) Crime distribution by area

d.) Crime occurrence by month/day/time

e.) Solved vs unsolved cases.

**<u>Techniques Used:</u>**

Descriptive statistics

Frequency analysis

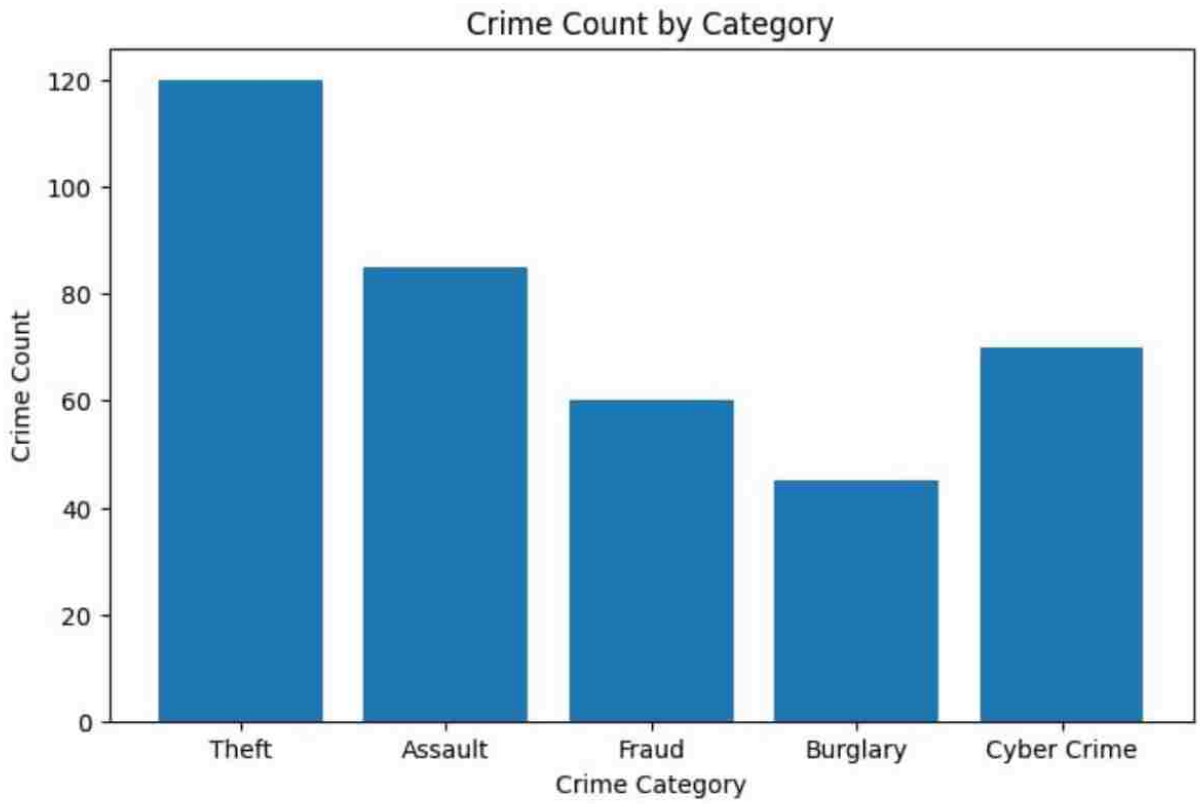
Grouping and filtering

# 6. Step 3 – Data Visualization

## A. Bar Chart – Crime Count by Category

**Purpose:**
To identify the most frequent crime categories.
**Example**:Theft,Assault
Burglary
Cyber Crime
**Insight:**Shows which crimes occur most frequently.

```
crime_counts = f['Crime_Type'].value_counts()

crime_counts.plot(kind='bar')
plt.title("Crime Count by Category")
plt.xlabel("Crime Type")
plt.ylabel("Count")
plt.show()
```




A. Bar Chart – Crime Count by Category

# B. Heatmap – Crime Hotspot Areas

## Purpose:

To identify high-crime locations.

## Visualization:

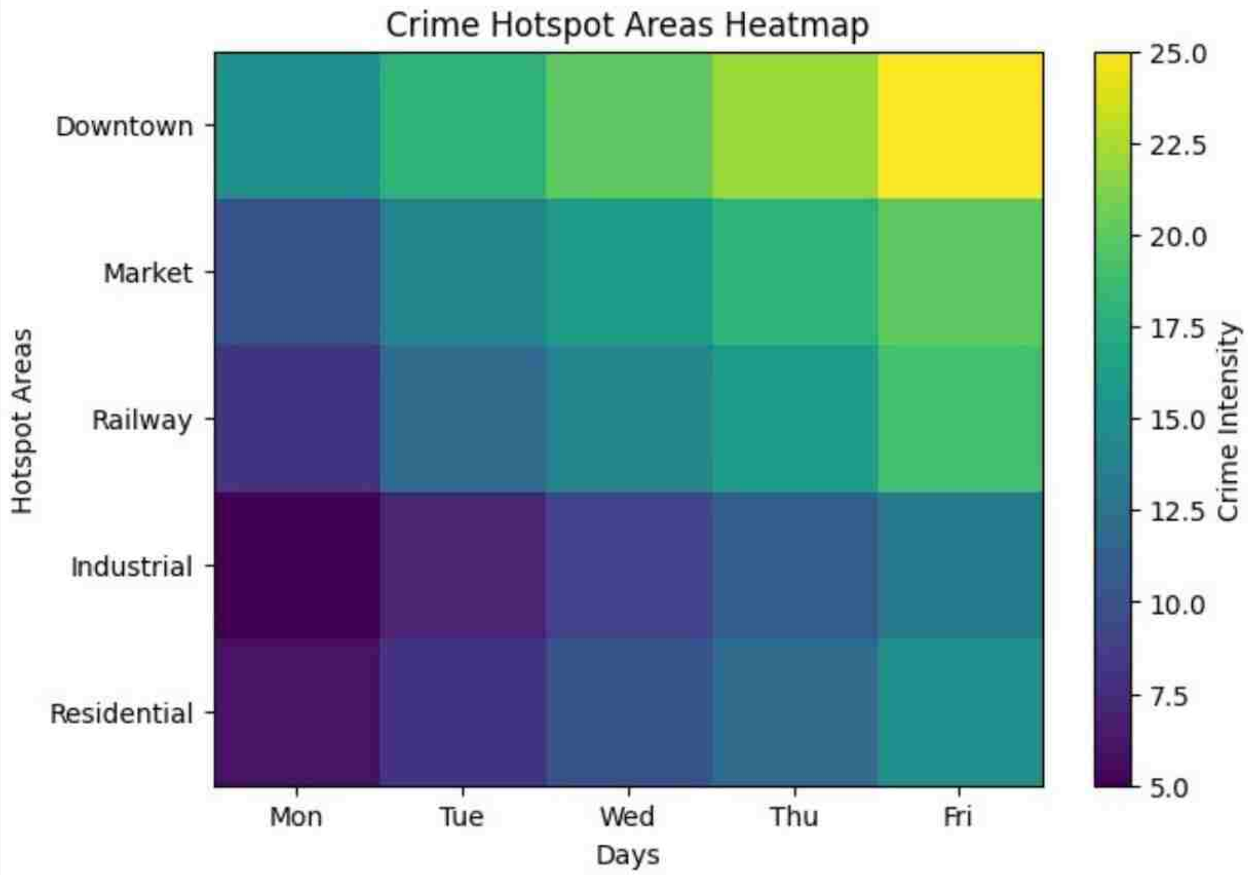
Geographic heatmap
Density-based hotspot map

## Insight:

Highlights unsafe areas with high crime concentration.

pip install matplotlib seaborn pandas

```python
import matplotlib.pyplot as plt
import seaborn as sns
import pandas as pd

# Data
crime_data = [
[15, 18, 20, 22, 25],  # Downtown
[10, 14, 16, 18, 20],  # Market
[8, 12, 14, 16, 19],   # Railway
[5, 7, 9, 11, 13],     # Industrial
[6, 8, 10, 12, 15]     # Residential
]

areas = ["Downtown", "Market", "Railway", "Industrial", "Residential"]
days = ["Mon", "Tue", "Wed", "Thu", "Fri"]

# Create DataFrame
df = pd.DataFrame(crime_data, index=areas, columns=days)

# Plot Heatmap
plt.figure(figsize=(8, 5))
sns.heatmap(
df,
cmap="viridis",
annot=False,
cbar_kws={"label": "Crime Intensity"}
)

plt.title("Crime Hotspot Areas Heatmap")
plt.xlabel("Days")
plt.ylabel("Hotspot Areas")

plt.show()
```




# B. Heatmap – Crime Hotspot Areas

# C. Line Chart – Crime Trends Over Time

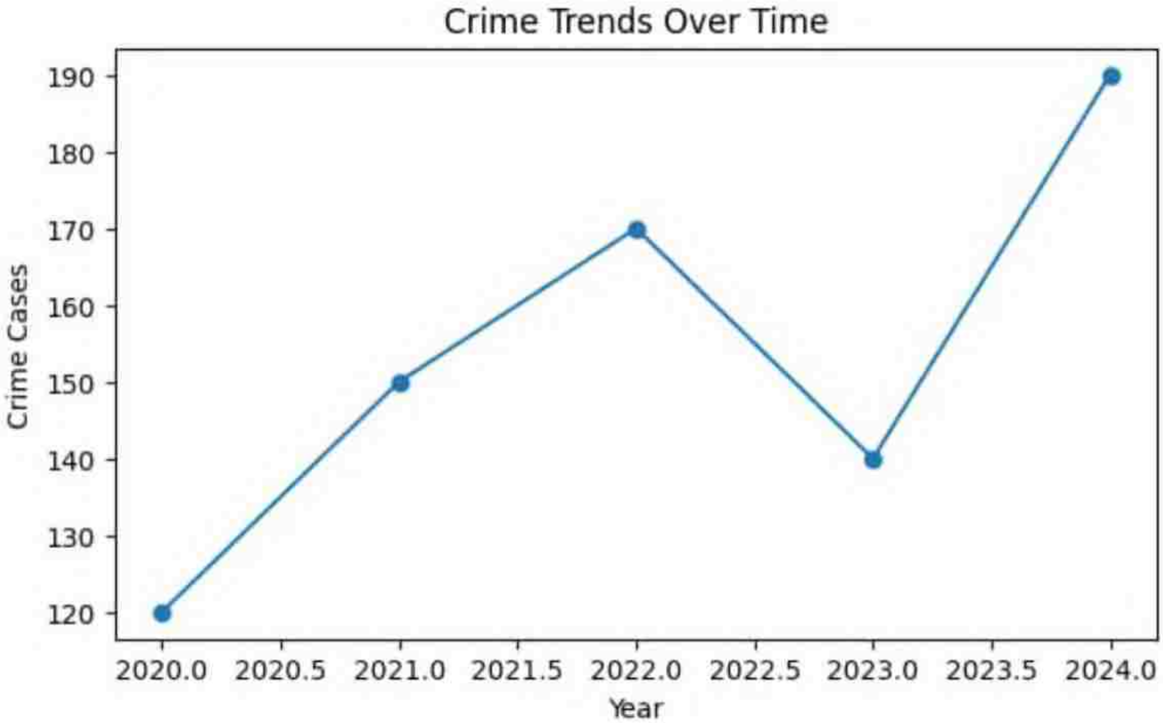
## Purpose:

To analyze crime occurrence trends monthly or yearly.

## Insight:

Shows whether crime rates are increasing or decreasing.

```
monthly_crimes = df.groupby(df['Date'].dt.month).size()

monthly_crimes.plot(kind='line', marker='o')
plt.title("Monthly Crime Trend")
plt.xlabel("Month")
plt.ylabel("Number of Crimes")
plt.show()
```




# C. Line Chart – Crime Trends Over Time

# D. Scatter Plot – Location vs Crime Frequency

## Purpose:

To analyze crime density across locations.

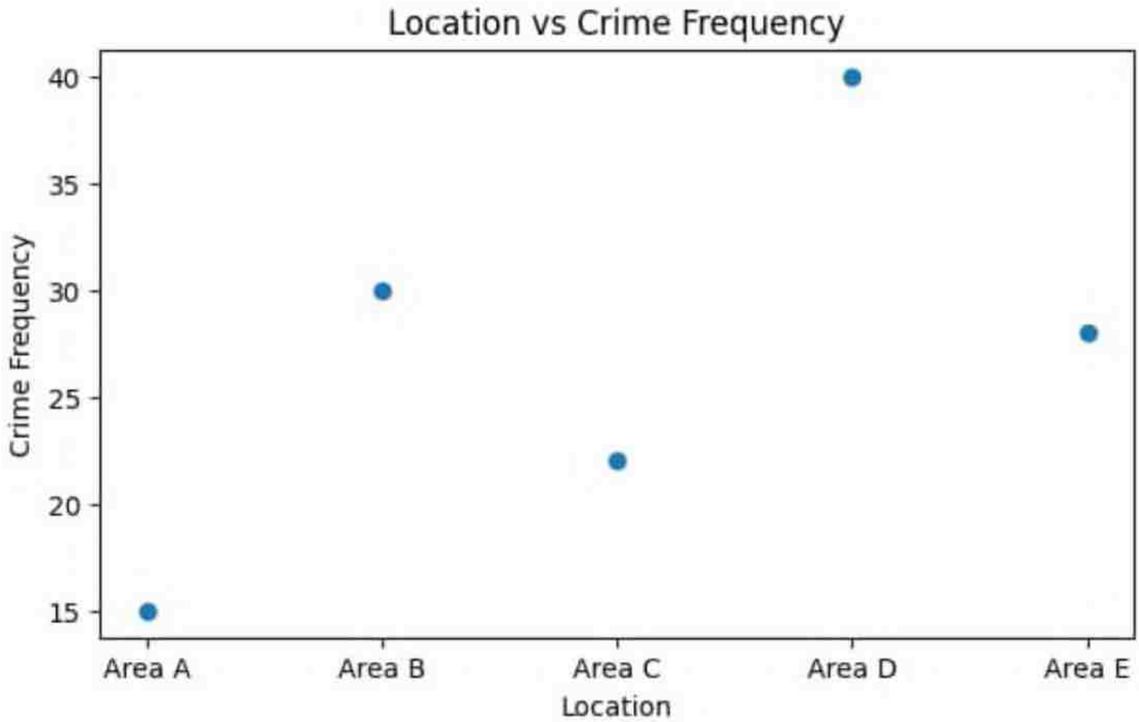
## Axes Example:

X-axis → Area/Location
Y-axis → Number of crimes

## Insight:

Shows highly affected regions.

```
location_counts = df['Location'].value_counts()

plt.scatter(location_counts.index[:20],
location_counts.values[:20])
plt.xticks(rotation=90)
plt.title("Crime Frequency by Location")
plt.show()
```




# D. Scatter Plot – Location vs Crime Frequency

# E. Box Plot – Crime Frequency Distribution by Crime Type

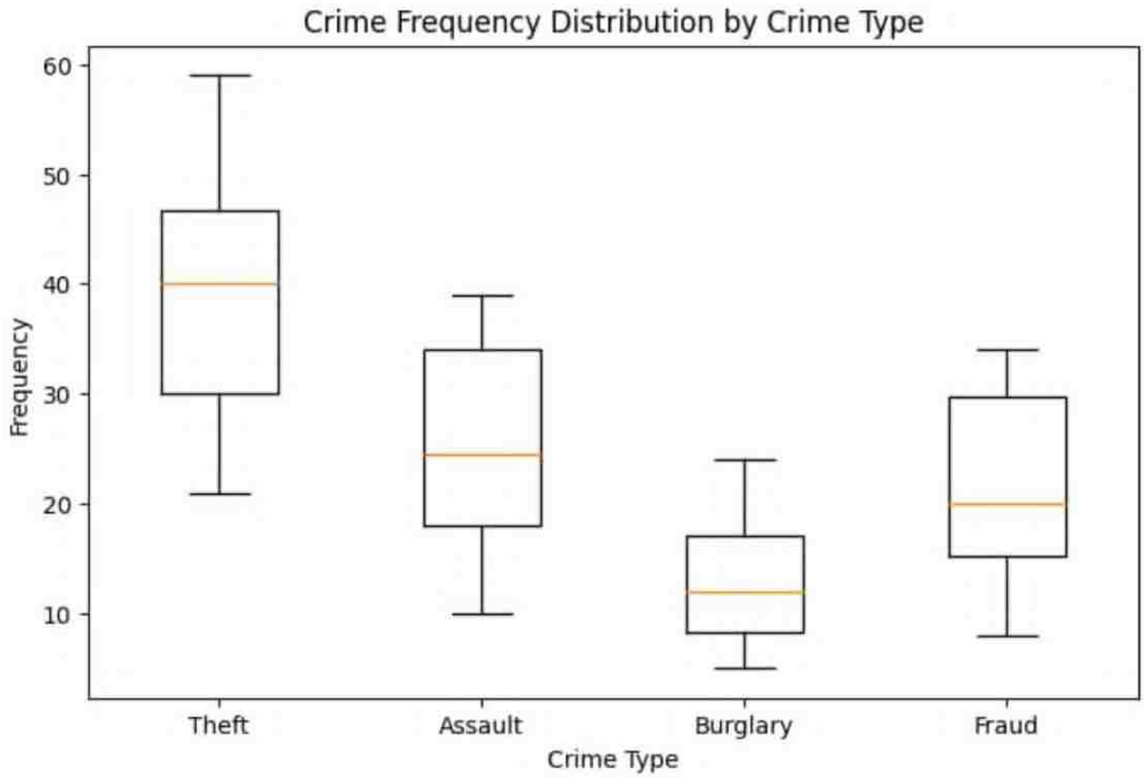
## Purpose:

To analyze variation in crime frequency across categories.

## Insight:

Identifies crime types with unusually high occurrences.

```python
sns.boxplot(x='Crime_Type', y='Crime_Count', data=df)

plt.xticks(rotation=90)
plt.title("Crime Distribution by Type")
plt.show()
```




# E. Box Plot – Crime Frequency Distribution by Crime Type

# F. Interactive Time-Based Crime Visualization

**Purpose:**

To visualize crimes based on:

A. time of day,
B. weekdays,
C.night vs daytime crimes.

**Insight:**

Identifies peak unsafe timings.

# G. Correlation Heatmap

## Purpose:

To analyze relationships between variables.

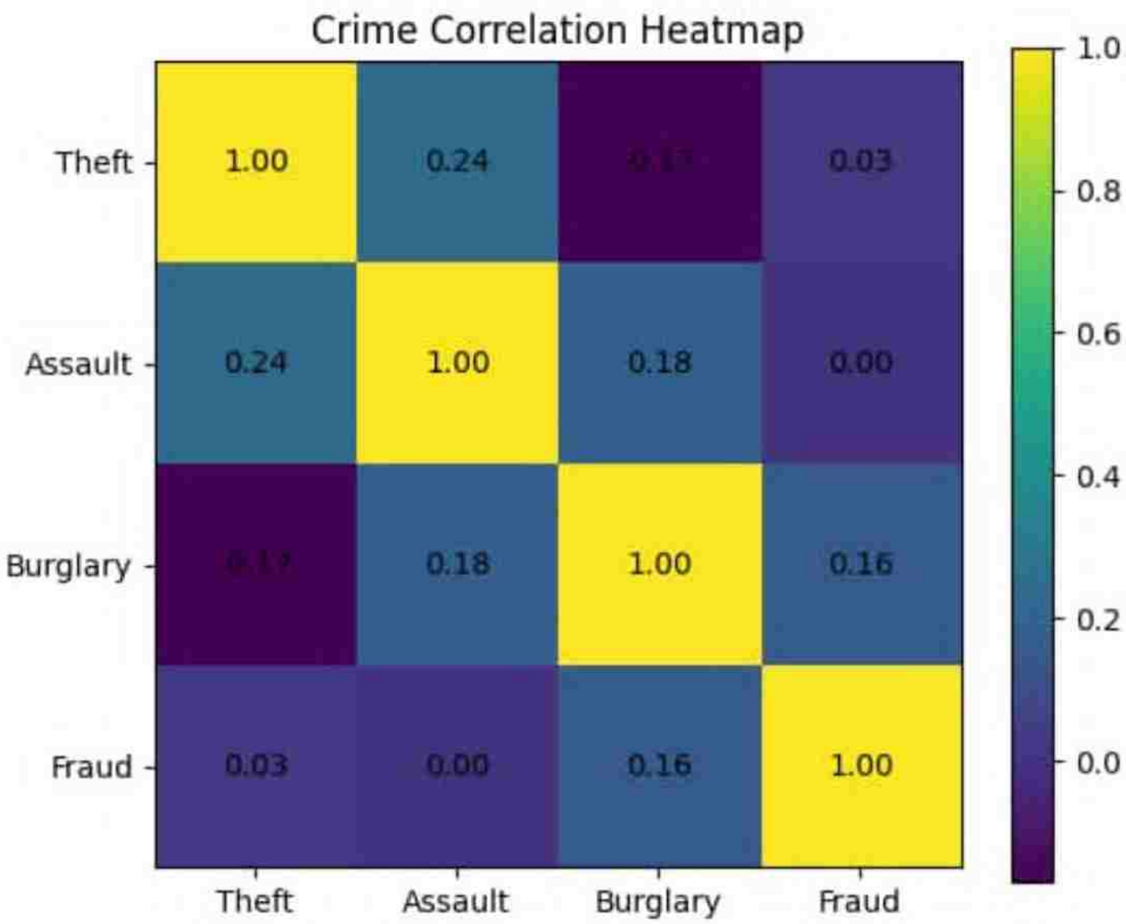
## Example Correlations:

Crime frequency vs time
Crime type vs location
Area population vs crime rate

## Insight:

Helps identify hidden patterns.

```
numeric_df = df.select_dtypes(include=['int64','float64'])

sns.heatmap(numeric_df.corr(), annot=True)

plt.title("Correlation Heatmap")
plt.show()
```




# Correlation Heatmap

# H. Geo-Spatial Crime Mapping

## Purpose:

To display crime locations on maps.

## Visualization:

Interactive maps
Marker clustering
Region-wise comparisons

# 7. Step 4 – Crime Hotspot Detection

**Purpose:**

To identify repeated crime zones.

**Techniques**:

Density analysis
Frequency clustering

**Output:**

High-risk areas
Frequently targeted zones

# 8. Step 5 – Key Insights

a. Theft is the most common crime category
b. Certain areas show consistently higher crime rates
c. Night-time crimes are more frequent
d. Some locations have low crime resolution rates
e. Crime frequency increases during weekends/festivals

# 9. Business Decisions / Recommendations

**Suggested Recommendations:**

Increase police patrols in hotspot areas

Install CCTV surveillance

Improve street lighting

Conduct public safety awareness campaigns

Increase monitoring during peak crime hours

# 10. Technologies Used

**Programming Language:**

Python

**Libraries:**

Pandas

NumPy

Matplotlib

Seaborn

Plotly

Folium

Visualization Tools:

Power BI

Tableau

# 11. Limitations

Incomplete crime records
Missing location coordinates
Limited geographic coverage
Data accuracy dependency

# 12. Future Scope

**Future Improvements:**

Real-time crime monitoring
AI-based crime prediction
Mobile safety applications
Smart city integration

# 13. Conclusion

1. This project successfully analyzed crime data using data visualization techniques to identify crime hotspots, major crime categories, and time-based crime trends.

2. Through visual analysis such as heatmaps,scatter plots, line charts, and correlation heatmaps, meaningful insights were obtained regarding unsafe locations and crime patterns.
3. The project demonstrates how data visualization can transform complex crime records into understandable insights that may supportbetter decision-making public awareness, and safety planning.

# Thank you

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